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Bike rack

Abstract

A bike rack for connecting to a connecting device of a vehicle to provide transportation functionality is provided. The bike rack includes a base; a carrying frame including at least one pair of carrying frame brackets that is configured to carry a front wheel and a rear wheel of a bike, each carrying frame bracket connected to the base, and a coupling portion arranged at one end of each carrying frame bracket distal to the base; a bearing mount located on one side of one of the at least one pair of carrying frame brackets; and a ramp board storable on the bearing mount. One end of the ramp board is formed with a corresponding coupling portion. The coupling portion of each carrying frame bracket is joinable with the corresponding coupling portion of the ramp board, such that the ramp board is inclined relative to the corresponding carrying frame bracket.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8272547	12/2011	Mital	224/501	B60R 9/10
11242101	12/2021	Kuschmeader	N/A	B60R 9/06
11383652	12/2021	Huang	N/A	B60R 9/10
11648885	12/2022	Wang	224/488	B60R 9/10
D1016723	12/2023	Wang	D12/407	N/A
D1045754	12/2023	Huo	D12/408	N/A
12187236	12/2024	Huo	N/A	B60R 9/10
D1068625	12/2024	Wang	D12/407	N/A
2010/0230455	12/2009	Wang	224/533	B60R 9/06
2020/0156724	12/2019	Kuschmeader	N/A	B60R 9/06
2020/0406825	12/2019	Wang	N/A	B60R 9/048
2022/0055703	12/2021	Kuschmeader et al.	N/A	N/A
2022/0203903	12/2021	Wang	N/A	B60R 9/06
2023/0202403	12/2022	Wang	224/549	B60R 9/06
2024/0010136	12/2023	Kuschmeader	N/A	B60R 9/10
2024/0253575	12/2023	Nilvius	N/A	B60R 9/06
2024/0351529	12/2023	Li	N/A	B60R 9/06
2024/0399972	12/2023	Willems	N/A	B60R 9/10
2025/0018873	12/2024	Weinert	N/A	B60R 9/06
2025/0083613	12/2024	Williams	N/A	B60R 9/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109689996	12/2018	CN	N/A
109720444	12/2018	CN	N/A
102012012889	12/2013	DE	N/A
202024107105	12/2024	DE	B60R 9/10
3756953	12/2019	EP	B60R 9/10
3756956	12/2019	EP	B60R 9/10
3978315	12/2021	EP	B60R 9/10
4019335	12/2021	EP	B60R 9/10
M473345	12/2013	TW	N/A
M593958	12/2019	TW	N/A
2023092035	12/2022	WO	N/A

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Background/Summary

FIELD

(1) The present disclosure generally relates to bike racks and, more particularly, to a bike rack, which is convenient for carrying bikes and storing ramp board.

BACKGROUND

(2) Racks are generally used for loading vehicles such as bikes, electric bikes, and electric scooters. A user must manually move the vehicle for loading or unloading during operation. When the weight of the vehicle is relatively heavy, part of the rack may be used in conjunction with a supporting board spanning between the ground and the rack, which allows the user to push the vehicle onto the rack. Thus, reducing labor.

(3) However, the previous supporting board relies directly on the rack, resulting in poor assembly stability. Additionally, the supporting board is an external attachment that hinders convenient storage and negatively impacts usability.

(4) Therefore, it is necessary to provide improvements addressing the shortcomings of well-known racks to resolve the aforementioned issues.

SUMMARY

(5) In a first aspect of the present disclosure, a bike rack for connecting to a connecting device of a vehicle to provide transportation functionality is provided. The bike rack includes: a base; a carrying frame including at least one pair of carrying frame brackets that is configured to carry a front wheel and a rear wheel of a bike, each carrying frame bracket of the at least one pair of carrying frame brackets connected to the base, and a coupling portion arranged at one end of each carrying frame bracket distal to the base; a bearing mount located on one side of one of the at least one pair of carrying frame brackets; and a ramp board storable on the bearing mount. One end of the ramp board is formed with a corresponding coupling portion. The coupling portion of each carrying frame bracket is joinable with the corresponding coupling portion of the ramp board, such that the ramp board is inclined relative to a corresponding carrying frame bracket.

(6) In another implementation of the first aspect, the bearing mount includes a first supporting frame and a second supporting frame. The first supporting frame and the second supporting frame are positioned on a same side of the one of the carrying frame brackets where the bearing mount is located, and each of the first supporting frame and the second supporting frame includes a U-shaped accommodating opening for placing the ramp board.

(7) In another implementation of the first aspect, each of the first supporting frame and the second supporting frame is equipped with a fixing strap.

(8) In another implementation of the first aspect, the ramp board includes a first plate body and a second plate body.

(9) In another implementation of the first aspect, one end of each of the first plate body and the second plate body is formed with the corresponding coupling portion.

(10) In another implementation of the first aspect, another end of the first plate body distal to the corresponding coupling portion has a connecting portion, and another end of the second plate body distal to the corresponding coupling portion has a corresponding connecting portion, and the first plate body is connectable to the corresponding connecting portion of the second plate body through the connecting portion, such that the first plate body and the second plate body are connected to form an integral structure.

(11) In another implementation of the first aspect, each of the connecting portion and the corresponding connecting portion is equipped with a coupling shaft and a coupling groove. When the connecting portion is connected to the corresponding connecting portion, the coupling shaft of the connecting portion inserts into the coupling groove of the corresponding connecting portion, and the coupling shaft of the corresponding connecting portion inserts into the coupling groove of the connecting portion to form a stable connection between the first plate body and the second plate body.

(12) In another implementation of the first aspect, the coupling portion comprises a protrusion. The corresponding coupling portion comprises a through-hole, and the ramp board couples the protrusion of the corresponding carrying frame bracket through the through-hole to secure the ramp board on one end of the corresponding carrying frame bracket.
(13) In another implementation of the first aspect, the bearing mount is fixed, by a screw-locking mechanism, to the one side of the one of the carrying frame brackets where the bearing mount is located.
(14) In another implementation of the first aspect, each carrying frame bracket is equipped with a fastening strap for securing the front wheel or the rear wheel of the bike.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure will be better understood from the following detailed description read in light of the accompanying drawings, where:
(2) FIG. **1** illustrates a schematic diagram of a bike rack installed on a vehicle in accordance with an example implementation of the present disclosure.
(3) FIG. **2** illustrates an external schematic diagram of the bike rack in accordance with an example implementation of the present disclosure.
(4) FIG. **3** illustrates a schematic diagram of a structure of a ramp board and a bearing mount in accordance with an example implementation of the present disclosure.
(5) FIG. **4** illustrates a schematic diagram of an assembly of the ramp board in accordance with an example implementation of the present disclosure.
(6) FIG. **5** illustrates a structural schematic diagram of a connecting portion and a corresponding connecting portion of the ramp board in accordance with an example implementation of the present disclosure.
(7) FIG. **6** illustrates a schematic diagram of a combination of the ramp board and a carrying frame bracket in accordance with an example implementation of the present disclosure.
(8) FIG. **7** illustrates a schematic diagram of an implementation of the bike rack in accordance with an example implementation of the present disclosure.
DETAILED DESCRIPTION
(9) The following disclosure contains specific information pertaining to exemplary implementations in the present disclosure. The drawings in the present disclosure and their accompanying detailed disclosure are directed to merely exemplary implementations. However, the present disclosure is not limited to merely these exemplary implementations. Other variations and implementations of the present disclosure will occur to those skilled in the art. Unless noted otherwise, like or corresponding components among the figures may be indicated by like or corresponding reference numerals. Moreover, the drawings and illustrations in the present disclosure are generally not to scale and are not intended to correspond to actual relative dimensions.
(10) For the purposes of consistency and ease of understanding, like features are identified (although, in some examples, not shown) by numerals in the exemplary figures. However, the features in different implementations may be different in other respects, and thus shall not be narrowly confined to what is shown in the figures.
(11) The disclosure uses the phrases “in one implementation,” “in some implementations,” and so on, which may each refer to one or more of the same or different implementations. The term “coupled” is defined as connected, directly, or indirectly through intervening components, and is not necessarily limited to physical connections. The term “comprising” means “including, but not necessarily limited to;” it specifically indicates open-ended inclusion or membership in the so-

described combination, group, series, and the equivalent.
(12) Additionally, for the purposes of explanation and non-limitation, specific details, such as functional entities, techniques, protocols, standards, and the like, are set forth for providing an understanding of the described technology. In other examples, detailed disclosure of well-known methods, technologies, systems, architectures, and the like are omitted so as not to obscure the disclosure with unnecessary details.
(13) FIG. **1** illustrates a schematic diagram of a bike rack installed on a vehicle in accordance with an example implementation of the present disclosure. A bike rack **10** may be connected to a connecting device of a vehicle **11**. The bike rack **10** is typically suspended at the rear of the vehicle **11** for transporting bike(s).
(14) With reference to FIG. **2**. FIG. **2** illustrates an external schematic diagram of the bike rack in accordance with an example implementation of the present disclosure. The bike rack **10** mainly includes a base **101**, a carrying frame **102**, a bearing mount **103**, and a ramp board **104**. The carrying frame **102** includes at least a pair of carrying frame brackets (**1021**, **1021′**) for carrying the front wheel and rear wheel of a bike. The carrying frame brackets (**1021**, **1021′**) are individually connected to the base **101**. As shown in FIG. **2**, the carrying frame brackets (**1021**, **1021′**) are positioned relative to the front and rear of the base **101**, and a coupling portion **1022** is arranged at one end of each of the carrying frame brackets (**1021**, **1021′**) distal to the base **101**. In some implementation, each of the carrying frame brackets (**1021**, **1021′**) is equipped with a fastening strap **1023** for securing the front wheel or the rear wheel of the bike. When the front wheel and rear wheel of the bike is respectively placed on the carrying frame brackets (**1021**, **1021′**), the fastening strap **1023** may secure the bike's front wheel or the rear wheel to prevent the bike from detaching from the bike rack **10** during transportation.
(15) FIG. **3** illustrates a schematic diagram of a structure of the ramp board and a bearing mount in accordance with an example implementation of the present disclosure. The bearing mount **103** is positioned on one side of one of the carrying frame brackets **1021**. The ramp board **104** is storable on the bearing mount **103**. The bearing mount **103** is located on one side of one of the carrying frame brackets **1021** of the bike rack **10**. The illustration in this figure is for explanatory purposes only, and it does not limit the placement position of the bearing mount **103**, as explicitly stated. As shown in FIG. **3**, in some implementation, the bearing mount **103** includes a first supporting frame **1031** and a second supporting frame **1032**. The first supporting frame **1031** and the second supporting frame **1032** are spaced apart and arranged on the same side of the carrying frame bracket **1021**. Each of the first supporting frame **1031** and the second supporting frame **1032** has a U-shaped accommodating opening **1033** which is suitable for placing the ramp board **104**. In some implementation, each of the first supporting frame **1031** and the second supporting frame **1032** is equipped a fixing strap **1034**. When the ramp board **104** is placed on the first supporting frame **1031** and the second supporting frame **1032**, the fixing strap **1034** may secure the ramp board **104** to prevent the ramp board **104** from falling. Furthermore, the bearing mount **103** may be fixed, by various ways, such as screw-locking, structural snaps, and other fixing methods, to one side of the carrying frame bracket **1021** where the bearing mount **103** is located.
(16) FIG. **4** illustrates a schematic diagram of an assembly of the ramp board in accordance with an example implementation of the present disclosure. One end of the ramp board **104** is formed with a corresponding coupling portion **1041**. The coupling portion **1022** of the carrying frame bracket **1021** is coupled to the corresponding coupling portion **1041** of the ramp board **104**. In some implementations, the ramp board **104** includes a first plate body **1042** and a second plate body **1043**. The connection of the first plate body **1042** and the second plate body **1043** forms the body of the ramp board **104**. Since the ramp board **104** includes the first plate body **1042** and the second plate body **1043**, the first plate body **1042** and the second plate body **1043** can be disassembled and separated, thus allowing the first plate body **1042** and the second plate body **1043** to overlap for storage. Additionally, the ramp board **104** is designed as a foldable structure for compact storage.

The structural form of the ramp board **104** is not limited to a specific configuration.
(17) With reference to FIG. **4**, in some implementations, one end of each of the first plate body **1042** and the second plate body **1043** is formed with a corresponding coupling portion **1041**. When the first plate body **1042** and the second plate body **1043** are connected, both ends of the ramp board **104** have the corresponding coupling portion **1041**, thus allowing the ramp board **104** to provide convenient and quick attachment to the coupling portion **1022** of the carrying frame bracket **1021**. In some implementations, another end of the first plate body **1042** that is distal to the corresponding coupling portion **1041** has a connecting portion **1044**, and another end of the second plate body **1043** that is distal to the corresponding coupling portion **1041** has a corresponding connecting portion **1045**. The first plate body **1042** is connectable to the corresponding connecting portion **1045** of the second plate body **1043** through the connecting portion **1044**, such that the first plate body **1042** and the second plate body **1043** are connected to form an integral structure. In some implementations, each of the connecting portion **1044** and the corresponding connecting portion **1045** is equipped with a coupling shaft **1046** and a coupling groove **1047**. When the connecting portion **1044** is combined with the corresponding connecting portion **1045**, the coupling shaft **1046** of the connecting portion **1044** inserts into the coupling groove **1047** of the corresponding connecting portion **1045**, and the connecting shaft **1046** of the corresponding connecting portion **1045** inserts into the connecting groove **1047** of the connecting portion **1044**, thus ensuring a stable connection between the first plate body **1042** and the second plate body **1043**. After the connection between the first plate body **1042** and the second plate body **1043** is established, the ramp board **104** is formed. When a bike positioned onto the ramp board **104**, both the connection portion **1044** of the first plate body **1042** and the corresponding connecting portion **1045** of the second plate body **1043** provide excellent load-bearing capacity, thus preventing deformation of the ramp board **104**.
(18) FIG. **5** illustrates a structural schematic diagram of the connecting portion and corresponding connecting portion of the ramp board in accordance with an example implementation of the present disclosure. In some implementations, the coupling portion **1022** includes a protrusion **1024**, and the corresponding coupling portion **1041** includes a through-hole **1048**. The ramp board **104** couples the protrusion **1024** of the corresponding carrying frame bracket **1021** through the through-hole **1048**, thereby securing the ramp board **104** on one end of the corresponding carrying frame bracket **1021**.
(19) With reference to FIGS. **4** to **7**, FIG. **6** illustrates a schematic diagram of the combination of the ramp board and the carrying frame bracket in accordance with an example implementation of the present disclosure. FIG. **7** illustrates a schematic diagram of an implementation of the bike rack in accordance with an example implementation of the present disclosure. The implementations of the present disclosure are as follows: (1): Placing the bike on the bike rack. The ramp board **104** that is placed in the bearing mount **103** is taken out in advance. Further, the first plate body **1042** is connected to the corresponding connecting portion **1045** of the second plate body **1043** through the connecting portion **1044**, thus forming the body of the ramp board **104**. Then, the corresponding coupling portion **1041** of the ramp board **104** is combined and fixed with the coupling portion **1022** of the carrying frame **102**, thus resulting in the ramp board **104** inclined relative to the carrying frame bracket **1021** and to facilitate the bike to be pushed onto the carrying frame bracket **1021** from the ground through the ramp board **104**. (2): Storage of ramp board. Once the bike is pushed onto the carrying frame **102** and fixed, the ramp board **104** is detached from the carrying frame **102** by separating the corresponding coupling portion **1041** from the coupling portion **1022** of the carrying frame **102**. The first plate body **1042** and the second plate body **1043** of the ramp board **104** are then disassembled and separated. Then, the first plate body **1042** and the second plate body **1043** are arranged in an overlapped manner and are storable on the bearing mount **103**. As such, the bike rack **10** provides convenience in both the use and storage of the ramp board **104**, thus effectively addressing deficiencies in well-known portable racks.

(20) The embodiments shown and described above are only examples. Many details are often found in the art. Therefore, many such details are neither shown nor described. Even though numerous characteristics and advantages of the present disclosure have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the present disclosure is illustrative only, and changes may be made in the details. It will therefore be appreciated that the embodiment described above may be modified within the scope of the claims.

Claims

1. A bike rack for connecting to a connecting device of a vehicle to provide transportation functionality, the bike rack comprising: a base; a carrying frame comprising: at least one pair of carrying frame brackets, the at least one pair of carrying frame brackets configured to carry a front wheel and a rear wheel of a bike, each carrying frame bracket of the at least one pair of carrying frame brackets connected to the base; and a coupling portion arranged at one end of each carrying frame bracket distal to the base; a bearing mount located on one side of one of the at least one pair of carrying frame brackets; and a ramp board storable on the bearing mount, one end of the ramp board formed with a corresponding coupling portion, wherein the coupling portion of each carrying frame bracket is joinable with the corresponding coupling portion of the ramp board, such that the ramp board is inclined relative to a corresponding carrying frame bracket.
2. The bike rack according to claim 1, wherein the bearing mount comprises a first supporting frame and a second supporting frame, the first supporting frame and the second supporting frame are positioned on a same side of the one of the carrying frame brackets where the bearing mount is located, and each of the first supporting frame and the second supporting frame includes a U-shaped accommodating opening for placing the ramp board.
3. The bike rack according to claim 2, wherein each of the first supporting frame and the second supporting frame is equipped with a fixing strap.
4. The bike rack according to claim 2, wherein the bearing mount is fixed, by a screw-locking mechanism, to the one side of the one of the carrying frame brackets where the bearing mount is located.
5. The bike rack according to claim 1, wherein the ramp board comprises a first plate body and a second plate body.
6. The bike rack according to claim 5, wherein one end of each of the first plate body and the second plate body is formed with the corresponding coupling portion.
7. The bike rack according to claim 6, wherein another end of the first plate body distal to the corresponding coupling portion has a connecting portion, and another end of the second plate body distal to the corresponding coupling portion has a corresponding connecting portion, and the first plate body is connectable to the corresponding connecting portion of the second plate body through the connecting portion, such that the first plate body and the second plate body are connected to form an integral structure.
8. The bike rack according to claim 7, wherein each of the connecting portion and the corresponding connecting portion is equipped with a coupling shaft and a coupling groove, when the connecting portion is connected to the corresponding connecting portion, the coupling shaft of the connecting portion inserts into the coupling groove of the corresponding connecting portion, and the coupling shaft of the corresponding connecting portion inserts into the coupling groove of the connecting portion to form a stable connection between the first plate body and the second plate body.
9. The bike rack according to claim 1, wherein the coupling portion comprises a protrusion, the corresponding coupling portion comprises a through-hole, and the ramp board couples the protrusion of the corresponding carrying frame bracket through the through-hole to secure the ramp

board on one end of the corresponding carrying frame bracket.
10. The bike rack according to claim 1, wherein the bearing mount is fixed, by a screw-locking mechanism, to the one side of the one of the carrying frame brackets where the bearing mount is located.
11. The bike rack according to claim 1, wherein each carrying frame bracket is equipped with a fastening strap for securing the front wheel or the rear wheel of the bike.